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2. Name the three filtration barriers that solutes must come across as they move from plasma to the lumen of Bowman's capsule. What components of the blood are usually excluded by these layers?
3. What forces promote glomerular filtration? What forces opposes them? What is meant by net filtration pressure?
4. Identify the following structures and explain their significance in renal physiology?
   a. Juxtaglomerular apparatus
   b. Podocytes
   c. Sphincters in the bladder
   d. Renal cortex
5. In which segment of the nephron most of the re-absorption of substances takes place?
6. When a molecule or ion is reabsorbed from the lumen of the nephron, where does it go? If a solute is filtered and not reabsorbed from the tubule, where does it go?

7. Match each of the following substances with its mode of transportation in proximal tubular reabsorption.

a. Na^1	-	simple diffusion
b. Glucose	-	primary active transport
c. Urea	-	indirect active transport
d. Plasma	-	paracellular movement
e. proteins	-	facilitated diffusion
f. Water	-	endocytosis

8. Which segment is the site of secretion and regulated reabsorption of ions and pH homeostasis?
9. What solute is normally present in the body to estimate GFR in humans?
10. Which part of the autonomic nervous system is involved in micturation process?
11. Match the following terms.

a. α-receptor arteriole	-	afferent
b. Autoregulation	-	basal lamina
c. Bowman's capsule	-	capillary blood pressure
d. Capsule fluid pressure	-	colloid osmotic pressure
e. Glomerulus	-	GFR
f. Podocyte	-	JG cells
g. Vasoconstriction	-	plasma proteins Norepinepherine

12. If the afferent arteriole of the nephron constricts, what happens to the GFR in that nephron? If the efferent arteriole constricts what happens to the GFR in that nephron? Assume that no auto regulation takes place.
13. How is the process of micturition altered by toilet training?
14. Concentration of urine depends upon which part of the nephron
    a. Bowman's capsule
    b. length of Henle's loop
    c. P.C.T.
    d. net work of capillaries arising from glomerulus
15. If Henle's loop were absent from mammalian nephron, which one of the following is to be expected?
    a. There will be no urine formation
    b. There will be hardly any change in the quality and quantity of urine formed
    c. The urine will be more concentrated
    d. The urine will be more dilute
16. A person who is on a long hunger strike and is surviving only on water, will have
    a. Less amino acids in his urine